

to be put in, even before the success of the platform has been gauged, and if at the end of the day, the whole thing fizzles out, there is an almost certain chance that it will be impossible to recuperate the losses.

Keeping oneself restricted to established platforms such as Android and iOS might be one solution which many of us seem to be content with, but another problem of fragmentation within these platforms is a devil which has been putting a roadblock in mobile app development time and again. The problem is bigger with Android, because there are almost innumerable devices of varying screen sizes, screen resolutions and underlying architecture. This makes it impossible to make sure that your app runs on all of the devices in the market. Not only does it make debugging/testing difficult, but native apps tend to be less stable and more prone to bugs if they are run on less popular devices.

POSSIBLE SOLUTION?

A solution that has always been spoken about for a long time is HTML5. It's one of those technologies which is open and supported by all devices, and is easily accessible irrespective of all hardware form factors. One reason is because HTML has always been encapsulated from directly interacting with the hardware and runs inside a container or interface which forms a part of the browser. This allows the device to provide a user experience which works almost at par with most native applications in combination with other client side technologies such as JavaScript and CSS3. This results in an interface which is fluid, flexible and reasonably responsive at the same time. While the last point is being hugely debated among the industry, it is no doubt that performance is also a lot dependent on server-side architecture and the user base that the application is aiming for in its particular use-case. Most major social media organisations such as Facebook and LinkedIn have dumped HTML5 in favour of native application development, but companies such as Sencha have clearly demonstrated how a good and responsive interface can be developed if correctly implemented, as you might be able to see in the form of its app called FastBook. Some of the negatives are now solved with features like the HTML5 application cache and offline browsing which allows the HTML5 app to be made available offline. This not only reduces server load, but also reduces the processing that takes place on the device itself.

The major benefit for HTML5, i.e. cross platform compatibility results in much lesser costs for development teams and generally a lesser time taken to reach the market. Such apps, once developed, can then be packaged and made to run in a web view container on any platform, and the best part is that they are accessible very easily even from desktop browsers and places like the Chrome web store etc. While these benefits can mean a lot for the developers, it does not mean that HTML5 is the solution for every such problem. It has its own limitations and its own use cases where it can be satisfactorily used. In some cases it does make sense to dump native apps in the favour of HTML5, but sometimes there are of course things

DB and Web Storage have also made great progress in the way web applications are now developed. These allow you to store and retrieve persistent key value data in the web clients. Web workers are also now available which provide functionality akin to background services in any operating system. In short, these allow your web application to run scripts in the background that do not need to run in response to a user based interaction, as is the case with normal JavaScript. All in all, HTML5 now forms a huge set of features which is now no less limited in functionality as opposed to any other programming language.

One major problem however, which is plaguing the HTML5 marketplace until now, has been the extremely slow pace of standardisation where each browser or vendor has had its own implementation of HTML5 APIs and features. There are now a lot of modern frameworks/tools available to supplement this technology and with the support of leading giants like Apple, Google, Intel, Mozilla etc., the trend for better compatibility has indeed sped up, wherein HTML5 features are much closer to a formal specification by the W3C.

Keeping these developments in mind, Intel has decided to ride the bandwagon for the open web of HTML5. It has recently announced the availability of a rich set of tools including the Intel XDK, HTML5 App Porter Tool and many others. As stated earlier, it is widely accepted that the success of a platform, whether hardware or software, depends a lot on the availability of content. It is clear that Intel too is aware of this fact, and the chip manufacturer understands that it would be in its best interests to make as many apps available on their Intel based devices, as are possible.

INTRODUCING THE INTEL XDK

The Intel XDK can be best defined as a sort of cloud based IDE which supplements the HTML5 mobile app development. One of its main features is the emulation of a lot of devices for HTML5 based apps and even hybrid native apps. Some of these devices include



The XDK login screen

which could be handled better by native apps. In the end, it all depends on the developers and the end product that they have in their mind. Sometimes, a hybrid approach makes sense too.

Some of the main features that HTML5 brings to the table are the video and audio elements, which allow embedding of multimedia content without using plugins like the flash player. This is expectantly much more secure and is faster than any third party plugin. More features such as inbuilt geolocation support and the much more popular HTML5 canvas are things that have changed the way we ever looked at the web. Web Semantics too are improving with better support for otherwise niche technologies like microdata and microformats, which allow processing of machine readable data. In fact there are some many standardized and non-standardized features that it could be impossible to list them in one place. Some advanced features like Indexed